Investigation: SPRINGER SA (CVM code 10960), fiscal year 2015, as it was on 2016-04-30

Statistical signal based on public financial statements. Not an accusation, not investment advice.

Company

Sector: Máquinas, Equipamentos, Veículos e Peças (origin: FCA); group: non-financial.
nenhum código de negociação no FCA entregue até a data (nos dados abertos, o campo só vem preenchido a partir do FCA de 2018)

Document used

2015 DFP, version 1, filed on 2016-03-31; origin: open data; type: consolidated; original scale: MIL.
Versions known on the date: 1 (2016-03-31).
Document at the CVM: http://www.rad.cvm.gov.br/ENETCONSULTA/frmDownloadDocumento.aspx?CodigoInstituicao=1&NumeroSequencialDocumento=54871

Possible manipulation (engine v1, study 1)

Beneish M-score
- Value: −3.798 (index)
- Alert by the study 1 rule: no
- Rule: M > −1.78 (pre-registration, section 4)
- Percentile in sector and year: 16 (n = 16)
- Percentile in group and year: 11 (n = 358)
- The company's own history: median −1.804; Tukey range (1.5 × IQR): from −5.149 to 0.834; value inside the range (index; 5 fiscal years)
- Confidence in the data: medium

DSRI, days' sales in receivables index
- Value: 0.786 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 8 (n = 18)
- Percentile in group and year: 16 (n = 430)
- The company's own history: median 0.769; Tukey range (1.5 × IQR): from 0.305 to 1.414; value inside the range (index; 5 fiscal years)
- Confidence in the data: medium

GMI, gross margin index
- Value: 1.308 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 81 (n = 18)
- Percentile in group and year: 88 (n = 449)
- The company's own history: median 0.699; Tukey range (1.5 × IQR): from −0.378 to 2.218; value inside the range (index; 5 fiscal years)
- Confidence in the data: medium

AQI, asset quality index
- Value: 1.334 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 81 (n = 18)
- Percentile in group and year: 89 (n = 432)
- The company's own history: median 1.065; Tukey range (1.5 × IQR): from −0.561 to 2.359; value inside the range (index; 5 fiscal years)
- Confidence in the data: medium

SGI, sales growth index
Value: 0.466 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 8 (n = 18)
Percentile in group and year: 4 (n = 451)
The company's own history: median 0.974; Tukey range (1.5 × IQR): from 0.501 to 1.748; value below the range (index; 5 fiscal years)
Confidence in the data: medium

DEPI, depreciation index
Value: 3.267 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 97 (n = 16)
Percentile in group and year: 99 (n = 375)
The company's own history: median 0.943; Tukey range (1.5 × IQR): from 0.766 to 1.135; value above the range (index; 5 fiscal years)
Confidence in the data: medium

SGAI, SG&A expenses index
Value: 1.162 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 81 (n = 18)
Percentile in group and year: 75 (n = 439)
The company's own history: median 0.998; Tukey range (1.5 × IQR): from 0.361 to 1.899; value inside the range (index; 5 fiscal years)
Confidence in the data: medium

TATA, total accruals to total assets
Value: −0.264 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 14 (n = 18)
Percentile in group and year: 8 (n = 456)
The company's own history: median −0.002; Tukey range (1.5 × IQR): from −0.246 to 0.359; value below the range (index; 5 fiscal years)
Confidence in the data: medium

LVGI, leverage index
Value: 0.814 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 8 (n = 18)
Percentile in group and year: 7 (n = 452)
The company's own history: median 0.747; Tukey range (1.5 × IQR): from 0.172 to 1.643; value inside the range (index; 5 fiscal years)
Confidence in the data: medium

DPO jump
Value: 8.8 days
Alert by the study 1 rule: no
Rule: DPO change in the year's top decile among non-financial companies (≥ 90th percentile of computable values on the date: 37.3 days)
Percentile in sector and year: 81 (n = 18)
Percentile in group and year: 71 (n = 419)
The company's own history: median −3.3 days; Tukey range (1.5 × IQR): from −49.4 days to 29.5 days; value inside the range (5 fiscal years)
Confidence in the data: medium
Alternative explanations: Bargaining power with suppliers and a change in purchase mix

(decision of 2026-11-07)

Implied interest rate minus average CDI
Value: 20.17 p.p.
Alert by the study 1 rule: yes
Rule: implied interest rate above the year's average CDI rate + 10 p.p. (pre-registration, section 4)
Percentile in sector and year: 64 (n = 18)
Percentile in group and year: 75 (n = 406)
The company's own history: median 39.32 p.p.; Tukey range (1.5 × IQR): from 4.73 p.p. to 68.24 p.p.; value inside the range (5 fiscal years)
Confidence in the data: medium
Alternative explanations: Expensive debt, exchange differences and lease interest inside finance costs (decision of 2026-11-07)

Accruals to average total assets
Value: −0.220 (index)
Alert by the study 1 rule: does not trigger alerts
Percentile in sector and year: 14 (n = 18)
Percentile in group and year: 12 (n = 455)
The company's own history: median −0.002; Tukey range (1.5 × IQR): from −0.225 to 0.326; value inside the range (index; 5 fiscal years)
Confidence in the data: medium

Financial risk (context only)

Altman Z" (financial risk, context only)
Value: 4.829 (index)
Components: X1 = 0.290; X2 = −0.116; X3 = −0.268; X4 = 4.864 (indices)
Percentile in sector and year: 87 (n = 19)
The company's own history: median 6.885; Tukey range (1.5 × IQR): from 1.445 to 10.530; value inside the range (index; 5 fiscal years)
- No cut-off zone: the available source does not define zones for Z"
- Coefficients estimated outside Brazil; bankruptcy models lose validity over time (Pereira and Martins 2015, p. 25)

Financial quality (context only)

Piotroski F (financial quality, context only)
Value: F = 2/9 (8 computable)
Signals:
  - Return on assets (ROA): 0
  - Operating cash flow on assets (CFO): 1
  - Change in ROA (ΔROA): 0
  - Cash above earnings (ACCRUAL): 1
  - Change in leverage (ΔLEVER): 0
  - Change in current ratio (ΔLIQUID): 0
  - No equity issuance (EQ_OFFER): —
  - Change in gross margin (ΔMARGIN): 0
  - Change in asset turnover (ΔTURN): 0
- No equity issuance (EQ_OFFER), from the cash flow statement: 6.03 lines of the frozen "emissões" pattern (reconnaissance, part 4.2), only those about shares or capital, as the article defines it ("did not issue common equity", p. 9); adaptation: preferred shares also count, because in Brazil they are equity, and the line is identified by text in the cash flow statement
- No long-term debt in either year: the change in leverage (ΔLEVER) is zero and the signal is

0 (project choice; the article only defines fell = 1, rose = 0, p. 8)

Changes

Restatement through the following year's comparative: no difference above the tolerance, or the next year's DFP did not exist yet on the date.

What was known on the date

- DFP 2010, v1: 2011-03-31
- DFP 2010, v2: 2011-04-19
- DFP 2011, v1: 2012-04-02
- DFP 2012, v1: 2013-03-27
- DFP 2013, v1: 2014-04-02
- FCA 2015, v1: 2015-03-05
- DFP 2014, v1: 2015-03-31
- FCA 2015, v2: 2015-05-26
- DFP 2015, v1: 2016-03-31

DFP and FCA filed with the CVM by the reference date; ITR and material facts come in later phases.

Data quality

53 applicable checks; 1 failed.

- Cash flow opening balance = cash and cash equivalents on the prior balance sheet (ÚLTIMO): difference above the tolerance of R$2,000

Auditor's report and auditor (factual section; no alert)

Auditor's report: Sem Ressalva (unqualified).

Excerpt checked literally: "Em nossa opinião, as demonstrações contábeis individuais acima referidas apresentam adequadamente, em todos os aspectos relevantes, a posição patrimonial e financeira da SPRINGER S/A. em 31 de dezembro de 2015, o desempenho de suas operações e os seus fluxos de caixa para o exercício findo naquela data, de acordo com as práticas contábeis adotadas no Brasil."

Auditor in the 2014 FCA: Confidor - Auditores Associados.

Auditor in the 2015 FCA: Confidor - Auditores Associados, Peppe Associados - Consultores & Auditores Independentes.

Auditor in the 2016 FCA: Peppe Associados - Consultores & Auditores Independentes.

Auditor changes (one line per change, canonical name by the firm's CVM code):

- 2015: Confidor - Auditores Associados -> Peppe Associados - Consultores & Auditores Independentes (transition: two firms in the 2015 FCA). The outgoing firm audited 5 consecutive fiscal years (2010 to 2014), by the start and end dates in the FCA. The change coincides with the end of the 5-fiscal-year term and is compatible with the rotation rule (art. 31 of CVM Instruction No. 308/1999, in force when the new firm started).

Auditor rotation: under CVM Resolution No. 23/2021, art. 31, the same independent auditor may not serve the same client for more than five consecutive fiscal years, with a minimum interval of three fiscal years before being hired again; under art. 31-A, the term is up to ten consecutive fiscal years when the company has a statutory audit committee in permanent operation and the auditor is a firm. The Resolution has been in force since 2021-04-01 (art. 41); before that, the equivalent rule was in arts. 31 and 31-A of CVM Instruction No. 308/1999, revoked by the Resolution (art. 40, I). A change of auditor is a fact, not an alert.

How the engine was tested

In study 1, engine v1 detected 7 of 13 detectable cases, with 108.3 false alarms per detected case over the whole period and an alert rate of 49.9% among evaluated non-financial company-years (docs/summary.md).

Limits

- No alert is not a clean bill of health. In Dechow, Sloan and Sweeney (1995), the most powerful

accruals model rejected 'no manipulation' for only 28.1% of 32 companies sanctioned by the SEC (Table 7, p. 32).
- In study 1, 49.9% of evaluated non-financial company-years (48.3% including financial companies) were in alert, and the number of detections did not beat chance in a descriptive test done after the result: 7 detections in 13 detectable cases, against 7.37 expected by chance without assuming independent years (8.56 assuming independent years).
- Risk models use numbers that may be manipulated (Pereira and Martins 2015, p. 29). That is why the three families are not added up.

Limitations

- Context indicators (Altman Z'' and Piotroski F) were not validated in Brazil by this project and do not trigger alerts

Evidence in the JSON: 38 DFP lines, each with document, version, filing date, account and location.

Engine

motor v1 (estudo 1), branch estudo-1-final, commit 8c9a823; chiocca.v2 (produto, fases B a E).
Generated at 2026-11-11T00:45:35+00:00.

Today's information (outside the reference date)

Registry status: CANCELADA. Source: CVM registry of listed companies (cad_cia_aberta.csv), in the version downloaded for the project, not on the reference date.
Name in today's registry: SPRINGER SA.